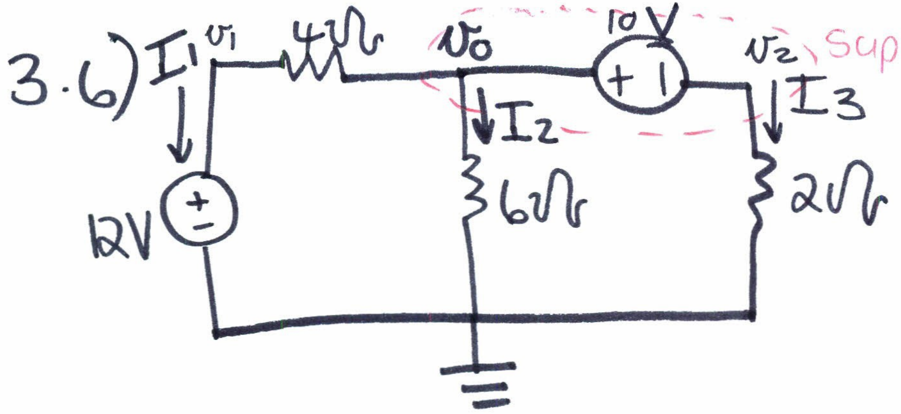
3.6)



$v_c = ?$

Supernode: Formed with voltage source between two non reference nodes

Nodal analysis: with Voltage sources.

Supernode: word gevorm met spannings bron tussen twee nie-verwysings nodes.

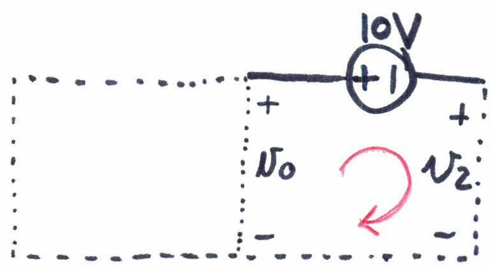
KCL: $I_1 + I_2 + I_3 = 0$

$$\frac{v_0 - v_1}{4} + \frac{v_0 - 0}{6} + \frac{v_2 - 0}{2} = 0$$

$$12(v_0 - v_1) + 8v_0 + 24v_2 = 0$$

$$20v_0 - 12v_1 + 24v_2 = 0 \quad \text{---} (1)$$

KVL op supernode:



$$-v_0 + 10 + v_2 = 0$$

$$v_0 = 10 + v_2$$

$$v_2 = v_0 - 10 \quad \text{----} (2)$$

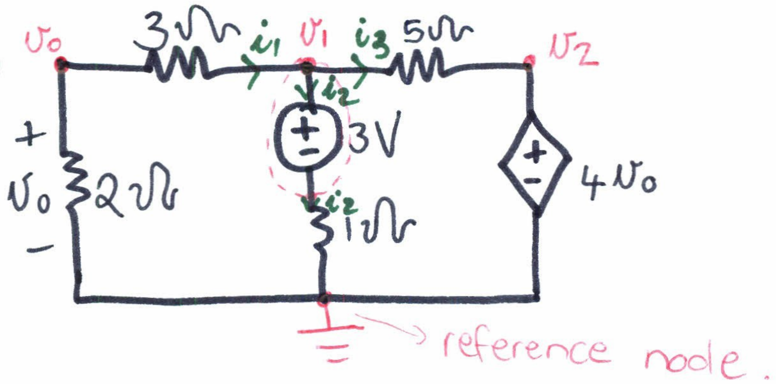
Substitute $\quad v_1 = 12\ V \quad$ ----- (3)

Vervang (3) & (2) in (1)

$$20v_0 - 12(12) + 24(v_0 - 10) = 0$$

$$44v_0 - 384 = 0$$

$$v_0 = 8.73\ V$$

3.8) 

KCL node v_1: $i_1 = i_2 + i_3$

$$\frac{v_0 - v_1}{3} = \frac{v_1 - 3}{1} + \frac{v_1 - v_2}{5}$$

$$5(v_0 - v_1) = \overset{15}{\cancel{3}}(v_1 - 3) + 3(v_1 - v_2)$$

$$5v_0 - 5v_1 = 15v_1 - 45 + 3v_1 - 3v_2$$

$$5v_0 = 23v_1 - 3v_2 - 45 \quad \text{----} \; (1)$$

$$v_2 = 4v_0 \quad \text{....} \; (2)$$

$$v_0 = \frac{2}{2+3} \cdot v_1 \quad \text{.....} \; (3)$$

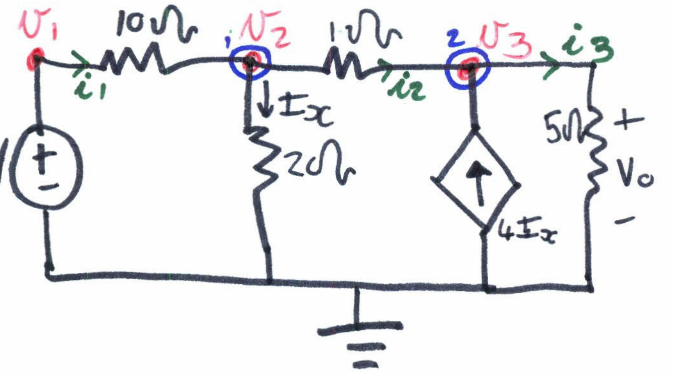
Voltage division (B| 44)

vervang 2&3 in (1)

$$5v_0 = 23\left(\frac{5}{2}v_0\right) - 3(4v_0) - 45$$

$$-40.5v_0 = -45$$

$$v_0 = \frac{10}{9} = 1.111 \text{ V}$$

3.12) 

V_o ?

KCL1: $i_1 - I_x - i_2 = 0$

$$\frac{v_1 - v_2}{10} - \frac{v_2 - 0}{2} - \frac{v_2 - v_3}{1} = 0$$

$$2(v_1 - v_2) - 10v_2 - 20(v_2 - v_3) = 0$$

$$2v_1 - 2v_2 - 10v_2 - 20v_2 + 20v_3 = 0$$

$$2v_1 - 32v_2 + 20v_3 = 0 \quad \cdots\cdots \; (1)$$

KCL2: $i_2 + 4I_x - i_3 = 0$

$$\frac{v_2 - v_3}{1} + 4I_x - \frac{V_o}{5} = 0$$

But!

Maar!

$v_1 = 30 \longrightarrow$

$v_3 = V_o \longrightarrow$

$I_x = \dfrac{v_2}{2} \longrightarrow$

$$5(v_2 - v_3) + 20I_x - V_o = 0$$

$$5v_2 - 5v_3 + 20I_x - V_o = 0 \quad \cdots\cdots \; (2)$$

$$5v_2 - 5V_o + 10v_2 - V_o = 0$$

$$15v_2 - 6V_o = 0$$

$$v_2 = \frac{6V_o}{15} \quad \cdots\cdots \; (3)$$

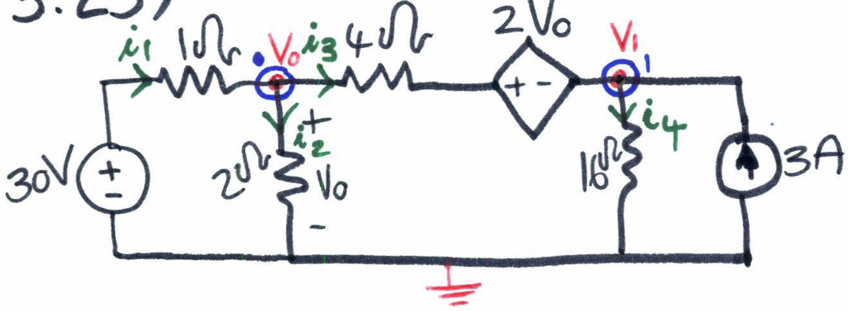
in (1)

$$60 - 32\left(\frac{6V_o}{15}\right) + 20V_o = 0$$

$$\frac{36}{5}V_o = -60$$

$$V_o = -8.33\dot{3}\ \text{V}$$

3.23)



KCL$_0$: $i_1 - i_3 - i_2 = 0$

$$\frac{30-V_0}{1} - \frac{V_0-(2V_0+V_1)}{4} - \frac{V_0-0}{2} = 0$$

$$8(30-V_0) - 2(V_0-(2V_0+V_1)) - 4V_0 = 0$$

$$240 - 8V_0 - 2(-V_0 - V_1) - 4V_0 = 0$$

$$-10V_0 + 2V_1 + 240 = 0$$

$$V_1 = 5V_0 - 120 \quad (\mathrm{V})$$

KCL1: $i_3 + 3 - i_4 = 0$

$$\frac{V_0-(2V_0+V_1)}{4} + 3 - \frac{V_1-0}{16} = 0$$

$$16(V_0-(2V_0+V_1)) + 192 - 4V_1 = 0$$

$$16(-V_0-V_1) + 192 - 4V_1 = 0$$

$$-16V_0 - 16V_1 - 4V_1 + 192 = 0$$

$$-16V_0 - 20V_1 + 192 = 0$$

$$-16V_0 - 20(5V_0-120) + 192 = 0$$

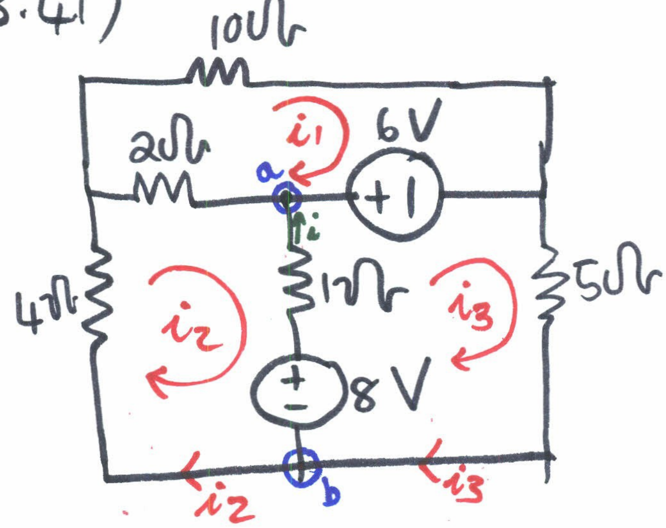
$$-16V_0 - 100V_0 + 2400 + 192 = 0$$

$$-116V_0 + 2592 = 0$$

$$-116V_0 = -2592$$

$$V_0 = 22.34\ \mathrm{V}$$

3.41)



KCL$_b$: $i + i_2 = i_3$

mesh is a loop without any other loops within.

For loop 1: $10i_1 - 6 + 2(i_1 - i_2) = 0$

$12i_1 - 2i_2 + 0i_3 = 6$

$6i_1 - i_2 + 0i_3 = 3$ ..... (1)

For loop 2: $4i_2 + 2(i_2 - i_1) + 1(i_2 - i_3) + 8 = 0$

$-2i_1 + 7i_2 - i_3 = -8$

$2i_1 - 7i_2 + i_3 = 8$ ..... (2)

For loop 3: $6 + 5i_3 - 8 + 1(i_3 - i_2) = 0$

$0i_1 - i_2 + 6i_3 = 2$ ..... (3)

Sit (1), (2) en (3) in matriks.

$$\begin{bmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \\ 2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 6 & -1 & 0 \\ 2 & -7 & 1 \\ 0 & -1 & 6 \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$\therefore \Delta = -234$

$$\Delta_2 = \begin{vmatrix} 6 & 3 & 0 \\ 2 & 8 & 1 \\ 0 & 2 & 6 \end{vmatrix} = 240$$

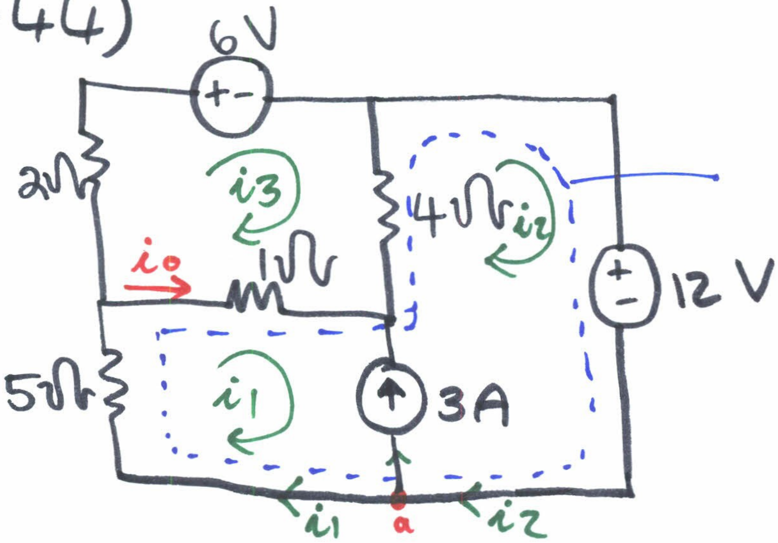
$$\Delta_3 = \begin{vmatrix} 6 & -1 & 3 \\ 2 & -7 & 8 \\ 0 & -1 & 2 \end{vmatrix} = -38$$

Vir i

$$i = i_3 - i_2$$

$$= \frac{\Delta_3 - \Delta_2}{\Delta} = \frac{-38 - 240}{-234} = 1.188\,A$$

3.44)



Soek i_o?

supermesh: if 2 meshes have a (dependent or independent current source in common.

For the supermesh:

$$12 + 5i_1 + 1(i_1 - i_3) + 4(i_3 - i_2) = 0$$

$$6i_1 - 4i_2 + 3i_3 = -12$$

$$-6i_1 + 4i_2 - 3i_3 = 12 \quad \ldots\ldots\ (1)$$

For loop 3: $+6 + 4(i_3 - i_2) + 1(i_3 - i_1) + 2i_3 = 0$

$$-i_1 - 4i_2 + 7i_3 = -6$$

$$i_1 + 4i_2 - 7i_3 = 6 \quad \ldots\ldots\ (2)$$

KCLa: $i_1 + 3 = i_2 \quad \ldots\ldots\ (3)$

vervang (3) in (1)

$$-6i_1 + 4(i_1 + 3) - 3i_3 = 12$$

$$-2i_1 + 12 - 3i_3 = 12$$

$$-2i_1 - 3i_3 = 0$$

$$2i_1 = -3i_3$$

$$i_1 = -\frac{3}{2} i_3 \quad \ldots\ (4)$$

ver vang (3) in (2)

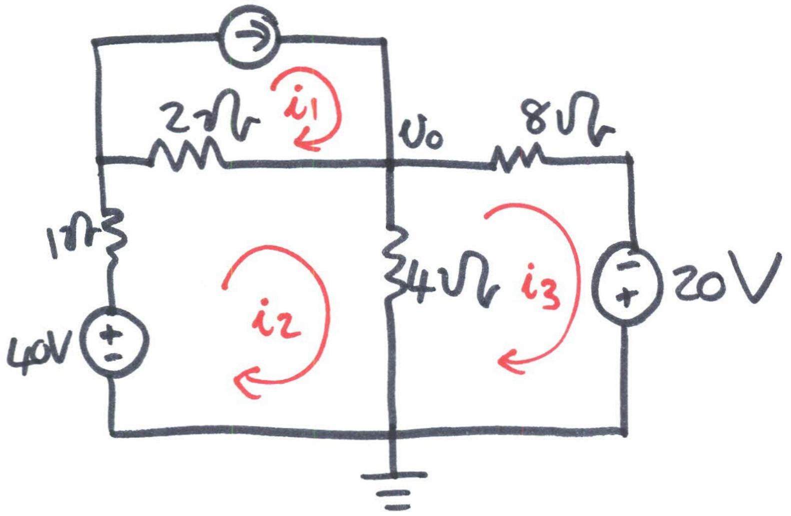
$$i_1 + 4(i_1 + 3) - 7i_3 = 6$$

$$5i_1 + 12 - 7i_3 = 6 \quad \ldots\ (5)$$

vervang (5) in (4)

$$5\left(-\frac{3}{2}\right)i_3 + 12 - 7i_3 = 6$$

3.51)



v_o?

8

loop 1: $i_1 = 5A$ .... (1)

loop 2: $-40 + i_2 + 2(i_2 - i_1) + 4(i_2 - i_3) = 0$

$-2i_1 + 7i_2 - 4i_3 - 40 = 0$ .... (2)

loop 3: $-20 + 4(i_3 - i_2) + 8i_3 = 0$

$-4i_2 + 12i_3 - 20 = 0$ .... (3)

Vervang (1) in (2)

$7i_2 - 4i_3 = 50$ ...... (5)

from (3) $-4i_2 + 12i_3 = 20$

$i_2 - 3i_3 = -5$

$i_2 = -5 + 3i_3$ .... (4)

Vervang (4) in (5)

$7(-5 + 3i_3) - 4i_3 = 50$

$-35 + 21i_3 - 4i_3 = 50$

$17i_3 = 85$

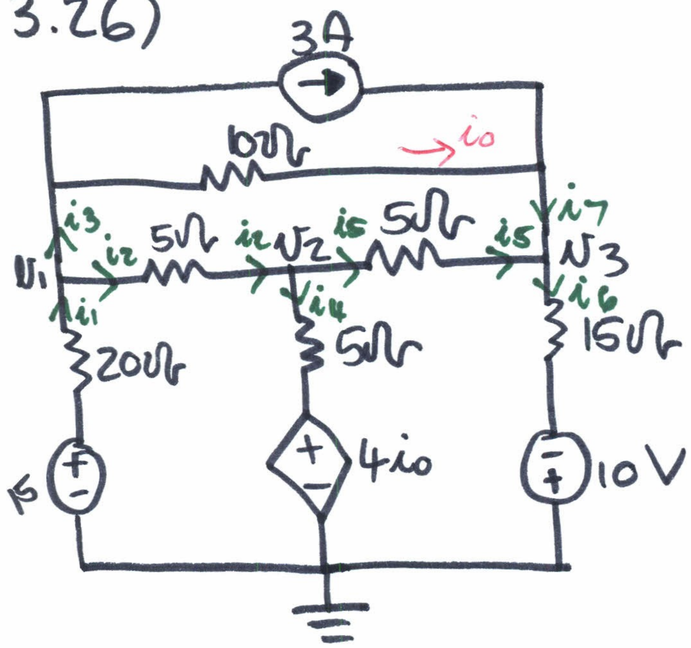
$i_3 = 5A$

$\therefore$ $i_2 = 10A$

$v_o = (-i_3 + i_2)(R)$

$= (-5 + 10)(4)$

3.26)



Soek V_1, V_2 & V_3?

Node 1: $i_1 = i_3 + i_2$

$$\frac{15 - V_1}{20} = \left(3 + \frac{V_1 - V_3}{10}\right) + \frac{V_1 - V_2}{5}$$

$$\therefore \quad 7V_1 - 4V_2 - 2V_3 = -45 \quad \ldots\ (1)$$

Node 2: $i_2 = i_4 + i_5$

$$\frac{V_1 - V_2}{5} = \frac{4i_o - V_2}{5} + \frac{V_2 - V_3}{5} \quad \ldots\ (2)$$

$$i_o = \frac{V_1 - V_3}{10} \quad \ldots\ (3)$$

ver vang (3) in (2)

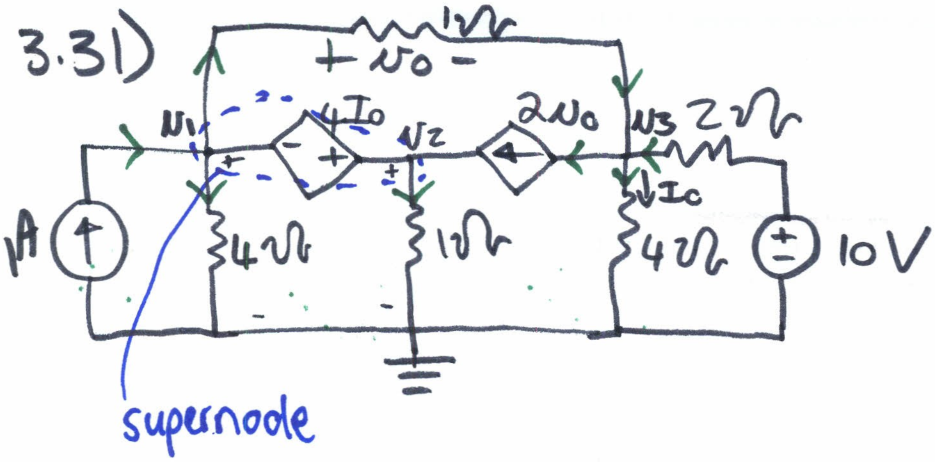
$$0 = 7V_1 - 15V_2 + 3V_3 \quad \ldots\ (4)$$

Node 3: $i_5 = i_6 - i_7$

$$\frac{V_2 - V_3}{5} = \frac{V_3 - (-10)}{15} + 3 + \frac{V_1 - V_3}{10}$$

$$\therefore \quad 70 = -3V_1 - 6V_2 + 11V_3 \quad \ldots\ (5)$$

3.31)



By die supernode:

$$1+2v_o = \frac{v_1}{4} + \frac{v_1 - v_3}{1} + \frac{v_2}{1} \quad \ldots ①$$

$$v_o = V_1 - V_3$$

$$\therefore ① \qquad 4 = -3v_1 + 4v_2 + 4v_3$$

By Node 3:

$$\frac{v_1 - v_3}{1} + \frac{10 - v_3}{2} = 2v_o + \frac{v_3}{4}$$

$$\therefore \quad 20 = 4V_1 + 0v_2 - V_3 \quad \ldots ②$$

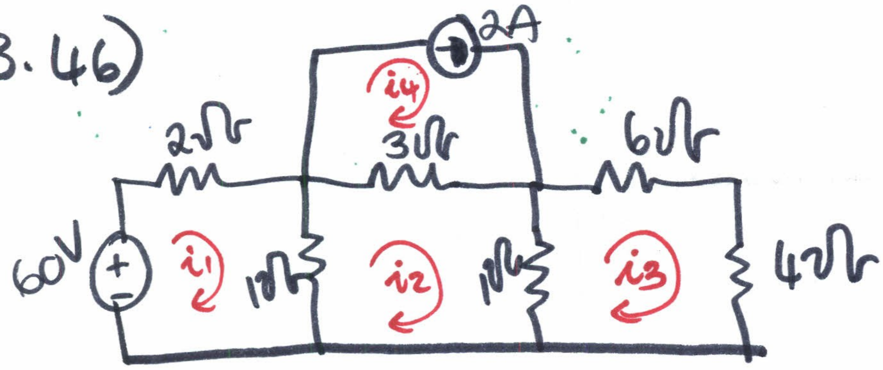
Supernode: $v_2 - v_1 - 4I_o = 0$

$$v_2 = v_1 + 4I_o$$

en $I_o = \frac{v_3}{4}$

$$\therefore v_2 = v_1 + v_3 \quad \ldots ③$$

3.46)



mesh analysis:

mesh1: $-60 + 2i_1 + 1(i_1 - i_2) = 0$

$3i_1 - i_2 = 60$ ..... ①

mesh 2: $1(i_2 - i_1) + 3(i_2 - i_4) + 1(i_2 - i_3) = 0$

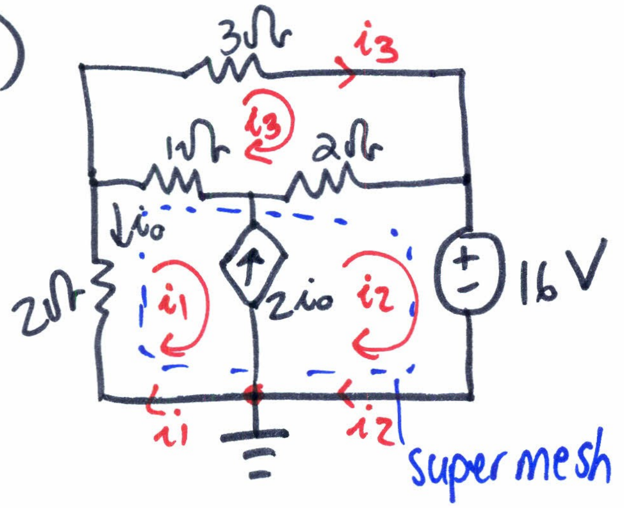
$-i_1 + 5i_2 - i_3 - 3i_4 = 0$ .... ②

mesh3: $10i_3 + 1(i_3 - i_2) = 0$

$-i_2 + 11i_3 = 0$ ..... ③

mesh4: $i_4 = 2A$ .... ④

3.49)



v_o?
i_o?

KCL ground: $i_1 + 2i_o = i_2$

For the supermesh:

$$2i_1 + 1(i_1 - i_3) + 2(i_2 - i_3) + 16 = 0$$

$$3i_1 + 2i_2 - 3i_3 + 16 = 0 \quad \dots (1)$$

at node $\perp$ $\quad i_1 + 2i_o = i_2$

& $\quad i_o = -i_1$

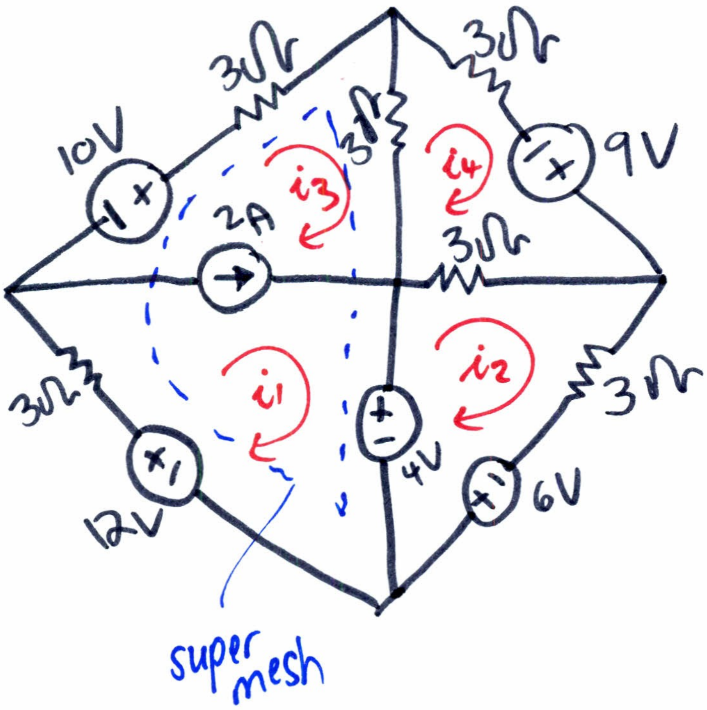
$$\therefore i_2 = -i_1 \quad \dots \dots (2)$$

loop 3: $\quad 3i_3 + 2(i_3 - i_2) + 1(i_3 - i_1) = 0$

$$-i_1 - 2i_2 + 6i_3 = 0$$

$$\therefore 6i_3 = -i_1 \quad \dots (3)$$

..54)



use mesh analysis:

$$i_1 - i_3 = 2 \quad \ldots \; (4)$$

From mesh (1) & (3):

$$-10 + 3i_3 + 3(i_3 - i_4) + 4 - 12 + 3i_1 = 0$$

$$3i_1 + 0i_2 + 6i_3 - 3i_4 - 18 = 0 \quad \ldots \; (1)$$

From mesh (2):

$$-4 + 3(i_2 - i_4) + 3i_2 - 6 = 0$$

$$6i_2 - 3i_4 - 10 = 0 \quad \ldots \; (2)$$

mesh 4:

$$-9 + 3(i_4 - i_2) + 3(i_4 - i_3) + 3(i_4) = 0$$

$$-3i_2 - 3i_3 + 9i_4 - 9 = 0 \quad \ldots \; (3)$$